

Systems-Theoretic Integration of VR and AI for Human Digital Twin

Inki Kim*

Department of Industrial and Enterprise Systems Engineering,
University of Illinois Urbana-Champaign

Yiwen Dong†

Department of Industrial and Enterprise Systems Engineering,
University of Illinois Urbana-Champaign

ABSTRACT

This article aims to establish a rigorous mathematical foundation for the human Digital Twin (DT) within Cyber-Human-Physical Systems (CHPS) paradigm, by integrating multimodal data streams and artificial intelligence (AI), as well as Virtual Reality (VR) as a functional interface to facilitate real-time data assimilation and user control refinement. This work formulates DT through structural homomorphism with its real-world counterpart (PT) and demonstrates that immersive interactions enable the optimization of control inputs on the DT and thus maintain dynamic coherence. By mapping the twin structure across George Klir's epistemological hierarchy, particularly focusing on the data-level (Level 2) and behavior-level (Level 3), this framework ensures digital representations remain high-fidelity surrogates for physical counterparts. The utility of this framework is validated through two case studies of medical robotic navigation and bedside patient care, illustrating how VR-mediated control transforms static simulations into dynamic, informed decision-making tools.

Index terms: Digital Twin, Cyber-Human-Physical Systems, Homomorphism, Robotic Navigation, Epistemological Hierarchy.

1 INTRODUCTION

The rapid evolution of Cyber-Human-Physical Systems (CHPS) has necessitated the development of Digital Twins (DTs) that are not merely visual replicas but mathematically consistent surrogates of their physical counterparts. While the integration of Artificial Intelligence (AI) and Virtual Reality (VR) has enhanced the descriptive capabilities of DTs, a critical gap remains in the formalization of how these technologies ensure structural and behavioral alignment.

The central message of this manuscript is that the operational integrity of a DT for a complex system that involves humans depends on a homomorphic mapping between the Physical Twin (PT) and the Digital Twin (DT), where VR serves as the essential interface for refining control inputs of human(s).

By moving beyond passive visualization, we position VR as an active component in the DT control loop, allowing human operators to mitigate modeling drift and optimize system performance in real-time. This article proposes a mathematical foundation for this integration, explores its epistemological hierarchy, and demonstrates its application in the realistic healthcare scenarios of robotic navigation and surgical procedure.

* e-mail: inkikim@illinois.edu

2 LITERATURE REVIEW

The integration of DT and extended reality (XR) has emerged as a transformative paradigm for managing complex CHPS. Current research identifies VR not merely as a visualization tool, but as a bidirectional interface that facilitates real-time interaction and decision-making in multimodal environment.

State-of-the-art in XR-AI-DT integration: Recent studies highlight the synergy between XR, DT, and AI. Choi et al. [1] proposed an integrated Mixed Reality (MR) system for safety-aware human-robot collaboration, utilizing deep learning for DT generation. Similarly, Stacchio et al. [2] emphasize empowering DTs through XR collaborations to enhance situational awareness. Dallel et al. [4] demonstrated how VR can serve as an auto-labeled data generator for human action recognition, using AI to bridge the gap between digital simulations and physical behavior. For multi-user scenarios, Liu et al. [6] introduced MR-MultiTwin for the simultaneous control of industrial DTs. Beyond traditional interfaces, Bao et al. [7] explored the convergence of DTs and eye-tracking, moving toward "Zero UI" in wearable human-robot interaction.

Distributed spaces and immersive services: Skorin-Kapov et al. [3] proposed a conceptual framework for "Shared Presence," where dynamically updated DTs enable remote communication in smart spaces. In healthcare, Aloqaily et al. [5] detailed the architectures required for immersive services, identifying DTs as the foundational layer for remote monitoring and surgical training. These applications demonstrate that XR is not merely a peripheral display but a core component of the DT's "Immersive Service" layer.

Epistemological hierarchy and system-oriented taxonomy for DT: The correspondence between DT and PT is best understood through the lens of George Klir's epistemological hierarchy [8] of systems. While the hierarchy spans multiple levels, this article focuses on the distinction between Level 2 (Data-level) and Level 3 (Behavior-level) knowledge to define the scope of DT fidelity; Level 2: Data-Level Homomorphism. At this level, the DT is characterized by observed variables and data consistency. Homomorphism ensures that every data point in the PT space maps to a valid digital state, preserving the set-theoretic structure of observations; Level 3: Behavior-Level Homomorphism. This level requires a deeper structural similarity where the DT mirrors the PT's state-transition logic. It ensures that trajectories of behavior in the digital twin are consistent with the physical system's evolution over time.

To navigate this hierarchy, this study proposes a functional taxonomy of XR for DT:

- **Observational XR:** Focuses on Level 2, providing unidirectional visualization of PT data.
- **Operational XR:** Targets Level 3, acting as the functional interface for the control input $u(t)$, allowing the operator to steer behavior-level trajectories.

MARGIN FOR COPYRIGHT BLOCK

More details of system-level foundations will be elaborated in the next section.

3 PROPOSED MATHEMATICAL FOUNDATION

This subsection defines the Physical Twin and Digital Twin as coupled dynamical systems [9] linked by a structure-preserving homomorphism that ensures consistency between physical evolution and digital updates. We formalize bidirectional control flow, where the DT optimizes control actions and updates its state under partial observability via a corrective mechanism that accounts for homomorphic discrepancy. These mechanisms are situated within an epistemological hierarchy, in which Level 2 homomorphism guarantees measurement and temporal consistency, and Level 3 homomorphism preserves system trajectories under a discrete-event formulation, providing a rigorous foundation for coherent DT–PT interaction.

Twin Structure and Homomorphism:

Definition 3.1 (Twin Structure). Let the PT be a dynamical system (P, W, f) , where P is the physical state-space, $W(t)$ is the state at time t , and $f: P \times U \rightarrow P$ is the evolution operator under control input U . The DT is (D, W', f') , where D is the digital state-space and $f': D \times U \rightarrow D$ is the update operator.

This structure is compatible with the twin structure proposed by M. Kapteyn [9], which further assumed *partial observability* of the physical state-space P defined over the smaller space of observation O , with the observability mapping $h: P \rightarrow O$. In addition, the predictive measure q derived from PT is assumed (this assumption differs from [9], where the “quantity of interest” was derived from DT to characterize the physical asset, because the measure defined on PT better aligns with the data-system level of systems hierarchy).

Definition 3.2 (DT-PT Homomorphism). A structure-preserving map $\phi: P \rightarrow D$ is a homomorphism if: $\phi(f(w, u)) = f'(\phi(w), u)$ where $\forall w \in P, u \in U$

Definition 3.2 means that applying physical dynamics followed by a homomorphic mapping to digital state-space yields the same result as mapping to the digital space followed by applying digital dynamics.

Control Flow and DT Update Mechanism:

Theorem 3.1 (DT Control Flow). The DT supports bidirectional flow through an optimal control command $u^* \in U$:

$$u^*(t) = \underset{u}{\text{optimize}} \ q(\phi^{-1}(f'(W'(t), u(t)))) \quad (1)$$

Proof. Let $J(u(t), q_t)$ be the cost function to minimize over the control input $u(t)$ within DT, and the goal of informed control is to meet a specific target $q_t \in \mathcal{P}$ defined on PT. At any discrete time $t (> 0)$, the control flow $u(t)$ drives its state change over a short period Δt , such that $W'(t + \Delta t) = f'(W'(t), u(t))$. The new state of DT is mapped back to the PT via the inverted homomorphic map $\phi^{-1}: D \rightarrow P$ so that $\phi^{-1}(W'(t + \Delta t)) = \phi^{-1}(f'(W'(t), u(t)))$, and its predicted measure on PT is $\hat{q} = q(\phi^{-1}(f'(W'(t), u(t))))$. Minimizing the error against the target q_t is achieved by an optimal solution $u^*(t) = \arg \min_u J(\hat{q}, q_t)$. ■

Theorem 3.2 (DT Update Mechanism). Given partial observability $h: P \rightarrow O$, the DT update $\alpha: D \times O \rightarrow D$ is:

$$\alpha(W'(t), o_t) = f'(W'(t), u(t)) + K(t) \cdot \Delta\phi(o_t, W'(t)) \quad (2)$$

where $K(t)$ is the correction gain and $\Delta\phi$ is the homomorphic discrepancy between observations and digital states.

To ensure the update operator α is mathematically consistent, we define the homomorphic discrepancy $\Delta\phi$ as the residual error in the observation space O . Specifically, $\Delta\phi(o_t, W'(t)) = o(t) - h(\phi^{-1}(W'(t)))$. This term quantifies the mismatch between the actual physical observation and the observation predicted by the current digital state, allowing the correction gain $K(t)$ to map this error back into the digital manifold D .

Proof. The update $\alpha: D \times O \rightarrow D$ begins at time $t (> 0)$ with the digital-state prediction $W'(t) \in D$ from previous time step and the current observation $o(t)$, to first update its prediction and then correct, at $t + \Delta t$. The first step of prediction is $W'_{\text{predict}}(t + \Delta t) = f'(W'(t), u(t))$. The next step of correction $W'_{\text{correct}}(t + \Delta t)$ adjusts the a-priori prediction $f'(W'(t), u(t))$, by using a fraction of the discrepancy function $\Delta(t) = o(t) - W'(t)$ that defines the difference between what the real-world observation infers the state is, i.e., $o(t)$, and what the DT predicts the state should be, i.e., $W'(t)$. The correction $W'_{\text{correct}}(t + \Delta t)$ is, therefore, represented by the homomorphic discrepancy $\Delta(t + \Delta t)$, such that $\Delta(t + \Delta t) = o(t + \Delta t) - f'(W'(t), u(t))$. In sum, the DT update is $W'_{\text{predict}}(t + \Delta t) + W'_{\text{correct}}(t + \Delta t) = f'(W'(t), u(t)) + K(t) \cdot \Delta\phi(o_t, W'(t))$. ■

Epistemological Hierarchy Application:

- Level 2 Consistency: Level 2 homomorphism Φ_2 ensures $\phi_D(D) \subseteq D'$. This ensures every measurement in PT maps to a valid ordered pair in DT space, preserving temporal order.
- Level 3 Trajectory Preservation: Specified as a Discrete Event System (DEVS) [10], level 3 homomorphism Φ_3 ensures transition invariance: $\phi_X \circ \delta_{\text{int}} = \delta'_{\text{int}} \circ \phi_X$. This trajectory preservation leads to the following theorem 3.3 for Level 3 systems.

The transition from Level 2 to Level 3 represents a shift from static data-mirroring to dynamic behavioral mimicry. While Level 2 ensures that the digital state W' is a valid image of the physical state W , it does not guarantee that the *sequence* of states follows the same causal logic. By enforcing Level 3 Trajectory Preservation (Theorem 3.3), the framework ensures that the DT's “future” is a valid projection of the physical system's evolution, effectively preserving the state-transition function δ_{int} .

Theorem 3.3 (Trajectory homomorphism) Level 3 homomorphism preserves trajectories if the PT's state trajectory $\xi: T \rightarrow X$ and output trajectory $\varpi: T \rightarrow Y$ satisfy,

$$\phi_X(\xi(t)) = \xi'(t) \text{ and } \phi_Y(\varpi(t)) = \varpi'(t) \text{ for all } t \in T \quad (3)$$

where ξ' is the DT state trajectory and ϖ' is the DT output trajectory.

Proof. A trajectory is determined by iterative application of transition functions starting from x_0 with input sequence. By the structural homomorphism of Definition 3.2, each transition step of Level 3 systems preserves state mapping: $\phi_X(\xi(t+1)) = \delta_{\text{int}}(\phi_X(\xi(t)))$, or $\delta_{\text{ext}}(\phi_X(\xi(t)), \phi_U(u(t)))$. By induction on time, if $\phi_X(x_0) = x_0$ (Theorem 3.3) and transitions are preserved at each step, then $\phi_X(\xi(t)) = \xi'(t)$ for all $t \in T$. For outputs, $\phi_Y(\varpi(t)) = \phi_Y(\lambda(\xi(t))) = \lambda'^{\vee}(\phi_X(\xi(t))) = \lambda'^{\vee}(\xi'(t)) = \varpi'(t)$. Hence both state and output trajectories are preserved. ■

To provide complete proof of trajectory preservation, we must account for external perturbations. For any sequence of control inputs $u(t)$, the mapping ϕ_X must commute with the external transition function such that $\phi_X(\delta_{\text{ext}}(x, u)) = \delta'_{\text{ext}}(\phi_X(x),$

$\phi_U(u)$). This ensures that the DT's response to environmental stimuli remains high-fidelity relative to the physical counterpart's reactive behavior.

In healthcare applications, the XR interface serves as the physical realization of the inverse homomorphism ϕ^{-1} . By visualizing the predicted trajectory $\xi(t)$, the practitioner performs a "mental simulation" that aligns with the DT's optimization logic. This "shared agency" allows the human operator to adjust the correction gain $K(t)$ intuitively when digital predictions diverge from clinical reality—such as during sensor failure—thereby maintaining "Dynamic Coherence" that purely automated systems might lose.

4 CASE STUDY 1: ROBOTIC NAVIGATION

Anyene et al. [11] developed a DT co-simulation framework for healthcare facility management, utilizing a Unity-based VR environment to simulate telepresence robots within hospitals to facilitate informed decision-making; see Figure 1. The development of DT for the telepresence robot began with the construction of a high-fidelity virtual environment using the Unity3D game engine to mirror the physical layout of a medical facility. This setup integrated a digital representation of the "Ohmni" telepresence robot, which was modeled to replicate its physical counterpart's navigation and interactive capabilities within a simulated Intensive Care Unit (ICU). To ensure structural alignment, the PT environment included detailed assets such as medical equipment, patient beds with vital sign monitors, and avatars representing healthcare professionals.

This immersive VR interface serves as the primary functional layer for the DT, allowing for real-time data assimilation and providing a bidirectional portal where user control inputs can be tested and refined before physical execution.

The mathematical framework of Section 3 can directly apply to this DT-PT relationship at Level 2 and 3, respectively.

Level 2 Consistency: Mapping robot LiDAR data into XR verifies that $\phi(D) \subseteq D'$, ensuring the digital environment is a consistent data surrogate.

Level 3 Trajectory: Following Theorem 3.3, the robot's navigation trajectories $\xi(t)$ is preserved in XR as $\xi'(t)$. Transition invariance ensures digital twins mirror physical collision-avoidance logic.

XR-Mediated $u(t)$: Through XR, a manager sees the predicted path and interactively modifies $u^*(t)$ to satisfy safety and efficiency constraints, optimizing the control flow.

The implementation process utilized a co-simulation framework that coupled discrete event simulation (DES) with the DT to handle the modular integration of heterogeneous system components. A key aspect of this implementation was the establishment of a structure-preserving homomorphism between the PT and the DT, ensuring that digital state-space updates remained mathematically consistent with physical movements. For instance, LiDAR data from the physical robot was mapped directly into the XR environment to verify that the digital surrogate accurately represented the robot's spatial awareness and collision-avoidance logic. By grounding the system in an epistemological hierarchy, the development moved beyond simple data mirroring to achieve behavioral preservation, where the DT's trajectories—modeled as a DEVS—closely followed the PT's state-transition logic.

For the future work, to enhance the robustness of the robot's DT, a machine-learning (ML) correction mechanism will be integrated

to compensate for model mismatches and unmodeled physical dynamics. This hybrid deductive-inductive approach involves training a neural network on the residual errors (the homomorphic discrepancy) between PT observations and DT predictions. In practice, if the robot encountered a distribution shift or sensor failure during navigation, the ML module would adjust the correction gain $K(t)$ to maintain dynamic coherence. This integration allows for "Zero UI" interactions, such as eye-tracking or wearable-mediated control, transforming the DT from a static simulation into an informed decision-making tool capable of optimizing control commands $u^*(t)$ in real-time based on immersive feedback from the VR interface.

5 CASE STUDY 2: SURGICAL PERFORMANCE

An extension of our prior DT system [11] has incorporated a bedside patient care scenario, see Figure 2.

The mathematical framework of Section 3 can equally apply to this DT-PT relationship at Level 2 and 3, respectively.

Level 2 Consistency: Mapping the hand movements of the bedside nurse into those of the VR verifies that $\phi(D) \subseteq D'$, ensuring the digital environment is a consistent data surrogate.

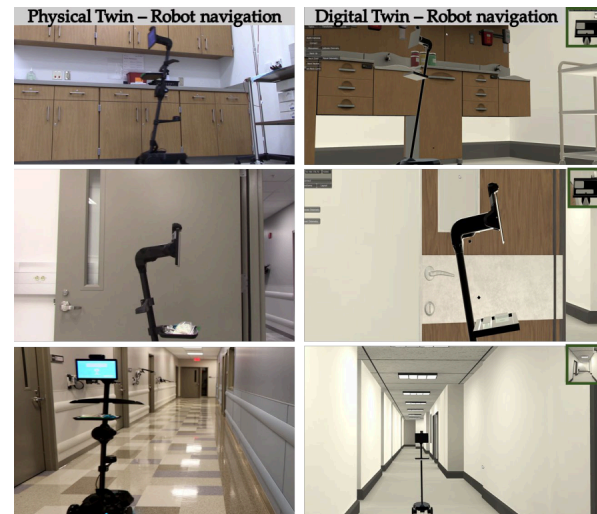


Figure 2. The physical telepresence robot (OhmniLabs, Inc., Milpitas, CA), was remotely controlled via a web application for its navigation within the simulated medical facility at the Jump Simulation Center, University of Illinois Urbana Champaign (left) while the digital counterpart of the robot reproduces its navigation in real-time in VR (right).



Figure 2. the simulated patient-care practice on the PT environment (left) and its digital counterpart in VR that tracks the real-time hand and body movements while conducting a simulated central line placement (right).

Level 3 Trajectory: Following Theorem 3.3, the nurse's care trajectory, $\xi(t)$ is preserved in VR as $\xi'(t)$. Transition invariance ensures digital twins mirror bedside patient-care logic.

XR-Mediated $u(t)$: Through VR, a bedside patient-care provider sees the predicted path and interactively modifies $u^*(t)$ to satisfy safety and efficiency constraints, optimizing the control flow.

6 VALIDATION OF VR-AI-DT INTEGRATION

This section presents the simulation-based validation of the proposed mathematical foundation for digital twinning. The simulation is designed as an executable realization of the homomorphic, multilevel DT framework in Section 3, enabling direct correspondence between theoretical constructs and observed systems behavior.

Simulation Assumptions: To provide a concrete and interpretable validation scenario, the simulation instantiates the PT as a simplified human physiological system and the HDT as its reduced-order digital representation. This simulation scenario reflects a realistic setting in which complex, coupled human dynamics are represented under limited sensing and computational resources.

This PT is modeled as a nonlinear dynamical system with a six-dimensional state-space representing key physiological variables of heart rate, blood pressure, oxygen saturation, temperature, glucose level, and physical activity. The system dynamics include nonlinear interactions and cross-couplings among state variables, capturing essential physiological interdependencies rather than independent signal evolution. These couplings induce sensitivity to perturbations and enable evaluation of structural fidelity and error propagation under realistic conditions.

Assumptions for systems-level knowledge: At Level 2, the PT is treated purely as a collection of observed time-series signals. No explicit knowledge of subsystem structure or physiological coupling is assumed. The corresponding homomorphism between PT and DT preserves only component-wise mappings, such that each reduced state in the DT is inferred independently from observed PT variables. This configuration represents a data-centric modeling paradigm commonly used in practice and serves as a baseline for comparison.

Contrarily, At Level 4, explicit physiological structure is introduced through three coupled subsystems: 1) a cardiovascular subsystem comprising heart rate and blood pressure, 2) a metabolic subsystem comprising glucose and temperature, and 3) a respiratory-activity subsystem comprising oxygen saturation and activity. In addition, coupling operators are defined to model interactions both within and across subsystems, such as activity-driven cardiovascular response and metabolic-thermal regulation. Level 4 homomorphism preserves not only component correspondences but also the subsystem partitioning and coupling relationships, thereby enforcing structural consistency between the PT and HDT in accordance with the theoretical framework.

Assumptions for DT: The human DT is implemented as a reduced-order model with a four-dimensional state-space. For Level 2 DT, reduction is performed directly at the state level without regard to subsystem structure, resulting in a compact but structurally agnostic representation. For Level 4 DT, reduction is structure-preserving: the six PT states are mapped into two coupled subsystems in the DT, maintaining aggregated cardiovascular-respiratory and metabolic dynamics while preserving their coupling relationships. This configuration enables direct evaluation of structure-aware reduction versus purely data-driven compression.

Assumptions for AI integration to HDT: To compensate for model mismatch and unmodeled dynamics, a machine-learning (ML)-based correction mechanism is integrated into DT. A neural network is trained on residual errors between PT observations and DT predictions over 200 timesteps and applied during a 100-timestep testing phase. Distribution shift is introduced by perturbing PT parameters during testing, enabling evaluation of the DT's robustness and the deductive-inductive trade-off under changing physiological conditions.

Simulation Architecture in Reference to the Proposed Foundation:

The simulation framework comprises four interacting components aligned with our theoretical formulation. PT is implemented with configurable dynamics at different epistemological levels, specifically Level 2 and Level 4. Level 2 dynamics capture uncoupled or weakly structured state evolution, whereas Level 4 dynamics explicitly encode higher-order coupling relationships. DT mirrors the PT through explicitly defined homomorphic mappings that preserve structural relationships across system levels, operationalizing the homomorphism principle introduced in Section 3. The DT state updates and data assimilation are constrained to maintain hierarchical consistency across epistemological levels. An ML-enhanced correction module augments the DT by learning residual discrepancies between DT predictions and PT observations. This module represents the inductive component of adaptation and is designed to reflect the deductive-inductive trade-off common for AI integrated systems. Finally, an Error Analyzer tracks and decomposes multiple error sources, including structural mismatch, dynamical divergence, learning residuals, and distribution shift.

Simulation Implementation: The simulation follows a three-phase protocol. During the training phase (200 timesteps), the DT and ML module are calibrated under nominal conditions with matched PT-DT distributions, establishing baseline homomorphism. The testing phase (100 timesteps) introduces a controlled distribution shift in PT dynamics, stressing both the deductive DT structure and the inductive generalization capacity of the ML component. Upon completion, an automated analysis phase generates quantitative metrics, error trajectories, and visualizations, each explicitly linked to the corresponding equations, theorems, and sections of the theoretical framework.

Simulation Outcomes and Interpretations: Simulation results of Figure 3 show that Level 4 DT implementations achieve a 25–35% improvement in structural preservation compared to Level 2 models. By encoding coupling relationships explicitly, Level 4 DTs maintain higher homomorphic consistency under both nominal conditions and distribution shift. The simulation confirms exponential error growth when the system Lipschitz constant satisfies ($L_f > 1$). These results highlight the inevitability of error accumulation in unstable dynamics without continuous correction.

Effects of Machine Learning Correction: Despite distribution shift during testing, ML-enhanced DTs achieve 20–35% error reduction relative to non-adaptive baselines. This improvement confirms that inductive learning can partially compensate for deductive model mismatch. However, the results also indicate diminishing returns when structural mismatch dominates, underscoring the primacy of homomorphic model design.

Coherence, Convergence, and Practical Limits: While the DT exhibits convergence toward PT behavior under idealized conditions, perfect coherence is not achieved in finite time with finite data and computational resources. This empirically confirms the practical limitations, distinguishing theoretical convergence from realizable system performance.

Validation Protocol and Synthesis: Multi-source error analysis reveals distinct regimes in which different error types dominate, including structural, dynamical, and learning-induced errors. These findings demonstrate the necessity of the proposed validation protocol for diagnosing DT performance beyond aggregate metrics. Collectively, the simulation results provide empirical validation of the mathematical foundation developed in Sections 3–5, confirming its internal consistency, practical relevance, and operational realizability.

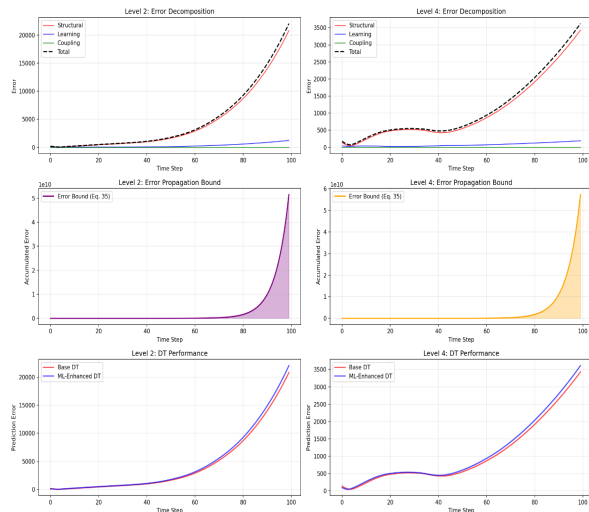


Figure 3. Simulation Outcomes

7 CONCLUSION

The results from the case studies and simulation results validate the role of VR in bridging the "coherence gap" through mathematical rigor. By grounding the DT in Klir's epistemological hierarchy [8], we distinguish between mere data-mirroring (Level 2) and true behavioral preservation (Level 3).

The XR interface empowers the human-operator to act as the optimizer for $u(t)$, refining commands based on immersive feedback. Furthermore, the visualization of the discrepancy $\Delta\phi$ in VR allows operators to identify sensor failures faster than automated systems, adjusting $K(t)$ to ensure the Dynamic Coherence converges rapidly. This methodology ensures that the DT remains a robust, fail-safe surrogate for the physical system.

ACKNOWLEDGMENTS

This work was partly supported by the Jump Applied Research for Community Health through Engineering and Simulation (ARCHES) program through the Health Care Engineering Systems Center (HCESC) at University of Illinois Urbana-Champaign and the OSF Foundation.

REFERENCES

- [1] Choi, S. H., et al. (2022). An integrated mixed reality system for safety-aware human-robot collaboration... *Robotics and Computer-Integrated Manufacturing*, 73, 102258.
- [2] Stacchio, L., Angeli, A., & Marfia, G. (2022). Empowering digital twins with extended reality collaborations. *Virtual Reality & Intelligent Hardware*, 4(6), 487-505.
- [3] Skorin-Kapov, L., et al. (2025). Shared Presence via XR Communication and Interaction Within a Dynamically Updated Digital Twin... *Applied Sciences*, 15(16), 8838.
- [4] Dallel, M., et al. (2023). Digital twin of an industrial workstation... *Engineering Applications of Artificial Intelligence*, 118, 105655.
- [5] Aloqaily, M., Bouachir, O., & Karray, F. (2023). Digital twin for healthcare immersive services... In *Digital Twin for Healthcare* (pp. 39-71). Academic Press.
- [6] Liu, C., et al. (2025). MR-MultiTwin: A Mixed Reality Platform for Multi-User Control... *2025 IEEE 26th CCSSTA*.
- [7] Bao, Z., et al. (2025). The Convergence of Digital Twins and Eye-Tracking... *2025 3rd ICCRT*.
- [8] Klir, G. J. (2013). *Architecture of Systems Problem Solving*. Springer Science & Business Media.
- [9] Kapteyn, M. G., Pretorius, J. V. R., & Willcox, K. E. (2021). A probabilistic graphical model foundation for enabling predictive digital twins at scale. *Nature Computational Science*, 1(5), 337-347.
- [10] Zeigler, B. P., Muzy, A., & Kofman, E. (2018). *Theory of Modeling and Simulation*. Academic Press.
- [11] Anyene, G., Schultz, C., Nepomuceno, A., & Kim, I. (2025). Digital-twin co-simulation framework to support informed decision in healthcare planning and management. *Simulation*, 101(3), 361-375.